MoonWalker Weekly Meeting

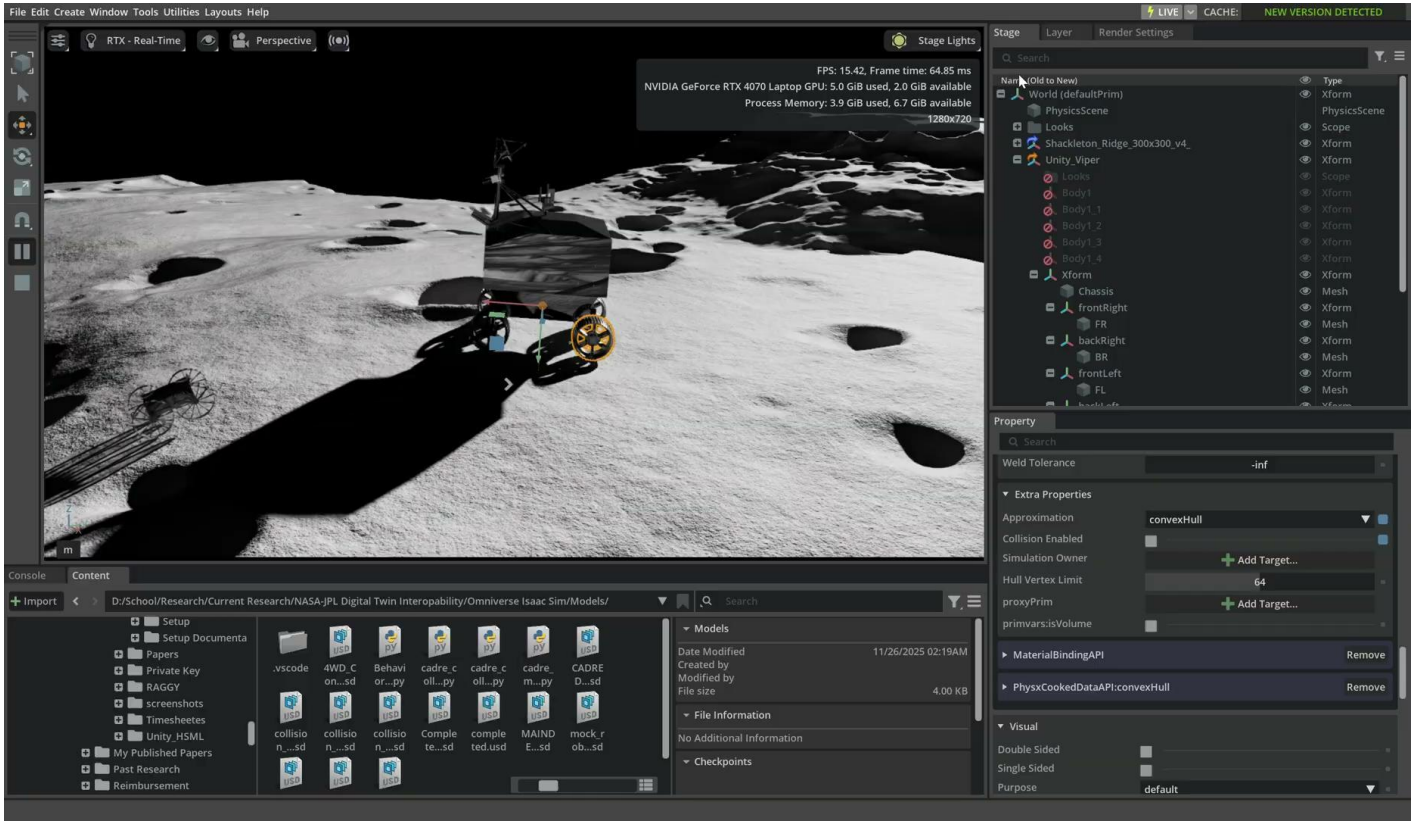
12/9 - 12/16 Summary

Elijah, Rowel, Martin

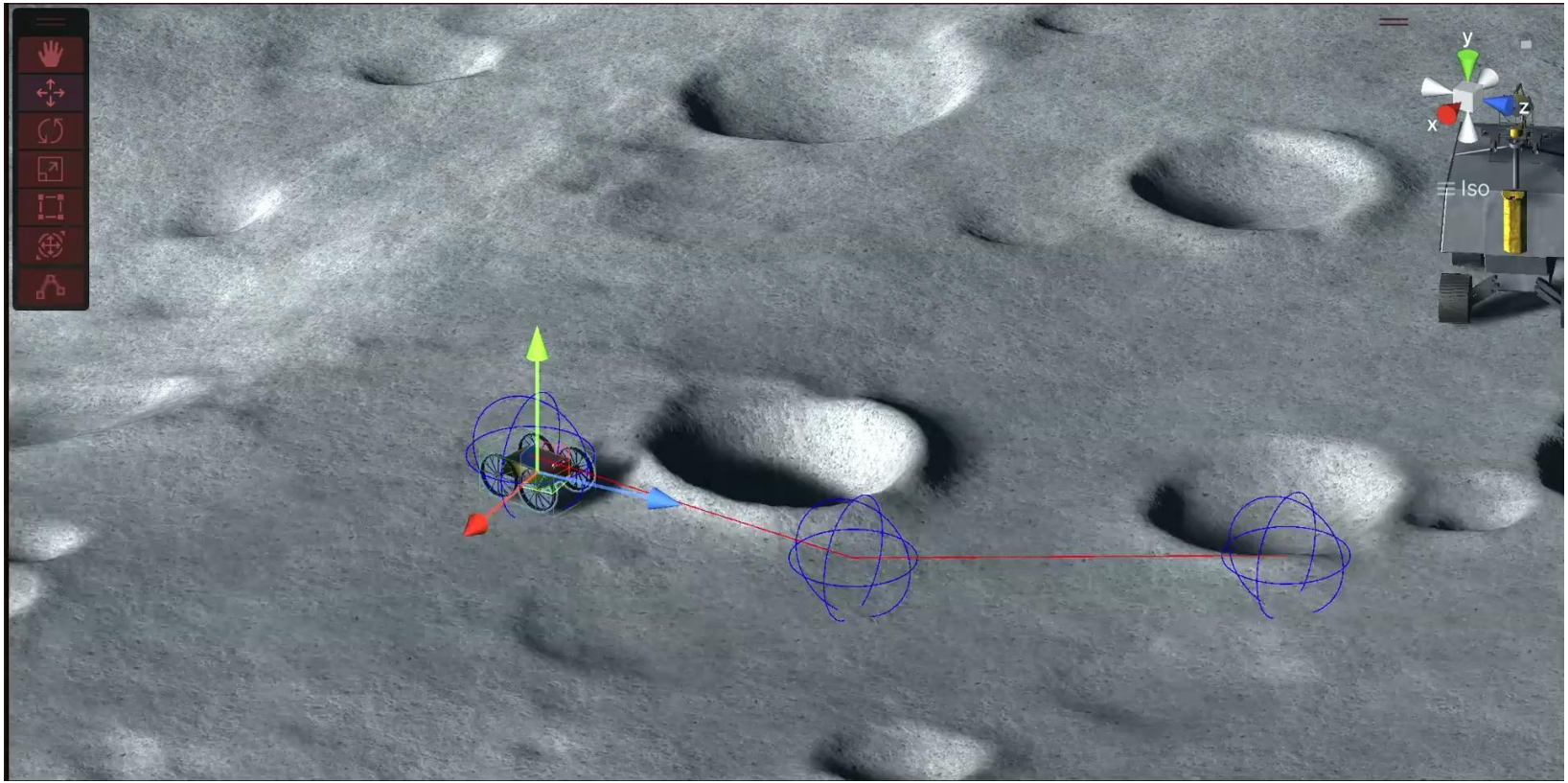
Progress Update

- Martin
  - Collision Testing with and without physics parameters (Shared)
  - Troubleshooting rover rotation/orientation issue
    - Figured out there’s a discrepancy between math systems used for Cadre rotations
    - Improving rotation stability by switching Unity Euler angles to Quaternions
  - Successful one-host handling and triggers on-collide
- Elijah
  - Collision Testing with and without physics parameters (Shared)
  - Troubleshooting rover rotation/orientation issue
  - Update scripts
    - Subscribe to Feedback Channel
    - Experiment with different message sending rates and physics parameters
- Rowel
  - Collision Testing with and without physics parameters (Shared)
  - Implement Feedback Channel
    - Aligned new topic access with existing validation
    - New authorization logic

Collision Test I with no Physics (Omniverse)

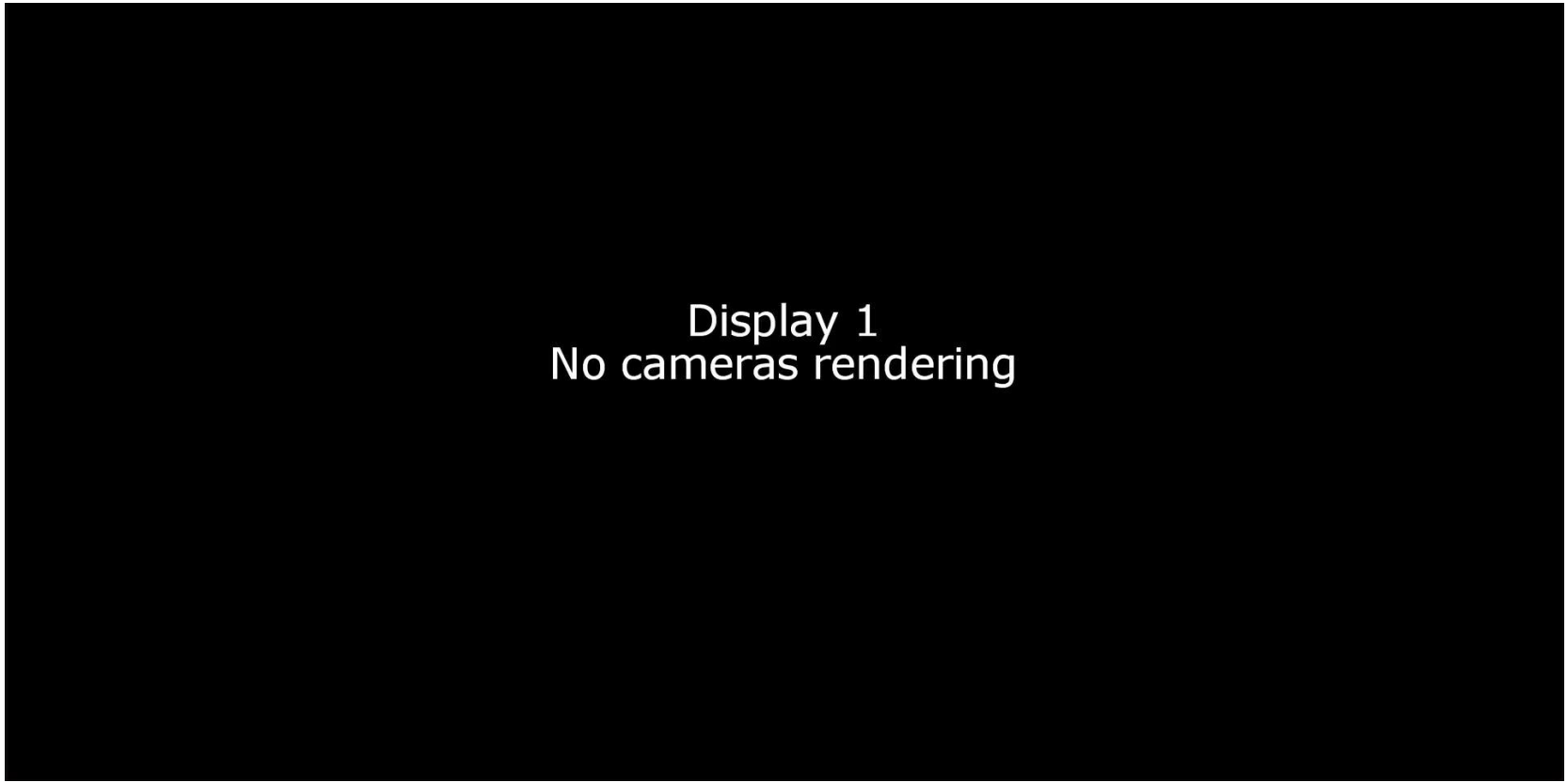


Collision Test I with no Physics (Unity)



Collision Test II with Physics (Omniverse): One-Host (Unity)

Collision Test II with Physics (Unity): One-host (Unity)



Action Plan

- Rowel
  - Potentially update the MySQL schema and Form-Website to provide more complex validation
  - Extend API to toggle between host roles
- Elijah
  - Finish fixing rotation/orientation problem with Unity
  - Extend and test with Omniverse as host
- Martin
  - Quaternions is relatively complicated to understand, researching it more for better rotations/orientation
  - Make one-host toggleable between simulations
  - Extend and test with Omniverse as host